EEG Emotion Recognition Using Dynamical Graph Convolutional Neural Networks

Tengfei Song, Wenming Zheng, *Member, IEEE*, Peng Song, *Member, IEEE*, and Zhen Cui

Abstract—In this paper, a multichannel EEG emotion recognition method based on a novel dynamical graph convolutional neural networks (DGCNN) is proposed. The basic idea of the proposed EEG emotion recognition method is to use a graph to model the multichannel EEG features and then perform EEG emotion classification based on this model. Different from the traditional graph convolutional neural networks (GCNN) methods, the proposed DGCNN method can dynamically learn the intrinsic relationship between different electroencephalogram (EEG) channels, represented by an adjacency matrix, via training a neural network so as to benefit for more discriminative EEG feature extraction. Then, the learned adjacency matrix is used to learn more discriminative features for improving the EEG emotion recognition. We conduct extensive experiments on the SJTU emotion EEG dataset (SEED) and DREAMER dataset. The experimental results demonstrate that the proposed method achieves better recognition performance than the state-of-the-art methods, in which the average recognition accuracy of 90.4 percent is achieved for subject dependent experiment while 79.95 percent for subject independent cross-validation one on the SEED database, and the average accuracies of 86.23, 84.54 and 85.02 percent are respectively obtained for valence, arousal and dominance classifications on the DREAMER database.

Index Terms—EEG emotion recognition, adjacency matrix, graph convolutional neural networks (GCNN), dynamical convolutional neural networks (DGCNN)

1 Introduction

Emotion recognition plays an important role in the human-machine interaction [1], which enables machine to perceive the emotional states of human beings so as to make machine more 'sympathetic' in the human-machine interaction. Basically, emotion recognition methods can be divided into two categories. The first one is based on non-physiological signals, such as facial expression images [2], [3], [4], [5], [6], body gesture [7], and voice signal [8]. The second one is based on physiological signal, such as electroencephalogram (EEG) [9], electromyogram (EMG) [10], and electrocardiogram (ECG) [11]. Among the various types of physiological signals, EEG signal is one of the most commonly used ones, which is directly captured from the brain cortex and hence it would be advantageous to reflect the mental states of human beings. With the rapid development of dry EEG electrode techniques and the EEG signal processing methods, EEG emotion recognition has received more and more attentions in recent years [12], [13], [14], [15].

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Basically, there are two major ways to describe human's emotions [16], i.e., the discrete basic emotion description approach and the dimension approach. For the discrete basic emotion description approach, the emotions are classified into a set of discrete status, e.g., the six basic emotions (i.e., joy, sadness, surprise, fear, anger, and disgust) [17]. Different from the discrete emotion description approach, the dimension approach describes emotions in continuous form, in which the emotions are characterized by three dimensions (valence, arousal and dominance) [18], [19] or simply two dimensions (valence and arousal), in which the valence dimension mainly characterizes how positive or negative the emotions are, whereas the arousal dimension aims to characterize the degree of how excited or apathetic the emotions are [16].

The research of applying EEG signal to the emotion recognition can be traced back to work of Musha et al. in [20]. During the past decades, many machine learning and signal processing methods are proposed to deal with the EEG emotion recognition [21], [22]. A typical EEG emotion recognition method usually consists of two major parts, i.e., discriminative EEG feature extraction and emotion classification. Basically, the EEG features used for emotion recognition can be generally divided into two kinds, i.e., time-domain feature type and frequency-domain feature type. The time domain features, e.g., Hjorth feature [23], fractal dimension feature [24] and higher order crossing feature [25], mainly capture the temporal information of EEG signals. Different from the time-domain feature, the frequency-domain feature aims to

使用动态图卷积神经网络进行脑电图情绪识别

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摘要—本文提出了一种基于新型动态图卷积神经网络（DGCNN）的多通道脑电图情绪识别方法。所提出的脑电图情绪识别方法的基本思想是使用图来对多通道脑电图特征进行建模，然后基于该模型进行脑电图情绪分类。与传统的图卷积神经网络（GCNN）方法不同，所提出的 DGCNN 方法可以通过训练神经网络动态地学习不同脑电图（EEG）通道之间的内在关系，以表示为邻接矩阵，从而有利于更具有判别性的脑电图特征提取。然后，使用学习到的邻接矩阵来学习更具有判别性的特征，以改进脑电图情绪识别。我们在 SJTU 情绪脑电图数据集（SEED）和 DREAMER 数据集上进行了广泛的实验。实验结果表明，所提出的方法比现有最先进的方法取得了更好的识别性能，其中在受试者依赖实验中实现了 90.4%的平均识别准确率，而 79.在 SEED 数据库上，独立交叉验证的准确率达到 95%，在 DREAMER 数据库上，效价、唤醒度和支配性分类的平均准确率分别为 86.23%、84.54%和 85.02%。

索引词—脑电图情绪识别，邻接矩阵，图卷积神经网络（GCNN），动态卷积神经网络（DGCNN）

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1 引言

情感识别在人机交互中扮演着重要角色[1]，它使机器能够感知人类的情感状态，从而使机器在与人交互时更加“富有同情心”。基本上，情感识别方法可以分为两类。第一类是基于非生理信号，例如面部表情图像[2][3][4][5][6]、身体姿态[7]和语音信号[8]。第二类是基于生理信号，例如脑电图（EEG）[9]、肌电图（EMG）[10]和心电图（ECG）[11]。在各种类型的生理信号中，脑电图（EEG）信号是最常用的之一，它直接从大脑皮层捕获，因此有利于反映人类的心理状态。

随着干电极脑电图技术和脑电图信号处理方法的快速发展，近年来脑电图情感识别受到了越来越多的关注[12][13][14][15]。

基本上，描述人类情绪主要有两种方法[16]，即离散基本情绪描述方法和维度方法。对于离散基本情绪描述方法，情绪被分类为一系列离散状态，例如六种基本情绪（即快乐、悲伤、惊讶、恐惧、愤怒和厌恶）[17]。与离散情绪描述方法不同，维度方法以连续形式描述情绪，其中情绪由三个维度（效价、唤醒度和支配度）[18]、[19]或简单地由两个维度（效价和唤醒度）来表征，其中效价维度主要表征情绪是积极还是消极，而唤醒度维度旨在表征情绪的兴奋程度或冷漠程度[16]。

将脑电图信号应用于情绪识别的研究可追溯至 Musha 等人的工作[20]。在过去的几十年里，许多机器学习和信号处理方法被提出用于处理脑电图情绪识别[21], [22]。典型的脑电图情绪识别方法通常由两个主要部分组成，即判别性脑电图特征提取和情绪分类。基本上，用于情绪识别的脑电图特征通常可分为两类，即时域特征类型和频域特征类型。时域特征，例如 Hjorth 特征[23]、分形维数特征[24]和高级交叉特征[25]，主要捕捉脑电图信号的时间信息。与时域特征不同，频域特征旨在

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capture the EEG emotion information from the frequency point of view. One of the most commonly used frequency-domain feature extraction methods is to decompose the EEG signal into several frequency bands, e.g., δ band (1-3 Hz), θ band (4-7 Hz), α band (8-13 Hz), β band (14-30 Hz) and γ band (31-50 Hz) [21], [26], [27], [28], and then extract EEG features from each frequency band, respectively. The commonly used EEG features include the differential entropy (DE) feature [29], [30], the power spectral density (PSD) feature [31], the differential asymmetry (DASM) feature [24], the rational asymmetry (RASM) feature [32] and the differential caudality (DCAU) feature [19].

To deal with EEG emotion classification problem, there are many methods appeared in the literatures [33], among which the method of using deep neural networks (DNN) [19] had been demonstrated to be one of the most successful one. Convolutional neural networks (CNN) is one of the most famous DNN approaches and had been widely used to cope with various classification problems, such as image classification [34], [35], [36], [37], object detection [38], tracking [39] and segmentation [40]. Although CNN model had been demonstrated to be very powerful in dealing with the classification problems, it is notable that previous applications of CNN focus more on the local feature learning from image, video and speech, in which the data points of the signal are continuously changed. For another feature learning problems, such as the feature learning from transportation network and brain network, the traditional CNN method may not be well suitable because the signals are discrete and discontinuous in the spatial domain. In this case, graph based description methods [41], [42] would provide a more appropriate way.

Graph neural network (GNN) [43] aims to build the neural networks under the graph theory to cope with the data in graph domain. Graph convolutional neural network (GCNN) [44] is an extension of the traditional CNN method by combining CNN with spectral theory [45]. Compared with classical CNN method, GCNN would be more advantageous in dealing with the discriminative feature extraction of signals in the discrete spatial domain [46]. More importantly, the GCNN method provides an effective way to describe the intrinsic relationship between different nodes of the graph, which would provide a potential way to explore the relationships among the multiple EEG channels during the EEG emotion recognition.

Motivated by the success of the GCNN model, in this paper we will investigate the multichannel EEG emotion recognition problem using graph representation approach, in which each EEG channel corresponds to a vertex node whereas the connection between two different vertex nodes corresponds to an edge of the graph. Although GCNN can be used to describe the connections among different nodes according to their spatial positions, we should predetermine the connections among the various EEG channels before applying it to building the emotion recognition model. On the other hand, it is notable that the spatial position connections among the EEG channels are different from the functional connections among them. In other words, a closer spatial relationship may not guarantee a closer functional relationship, whereas the functional relationship would be useful for the discriminative EEG feature extraction in emotion recognition. Consequently, it is not reasonable to predetermine the connections of the graph nodes according to their spatial positions.

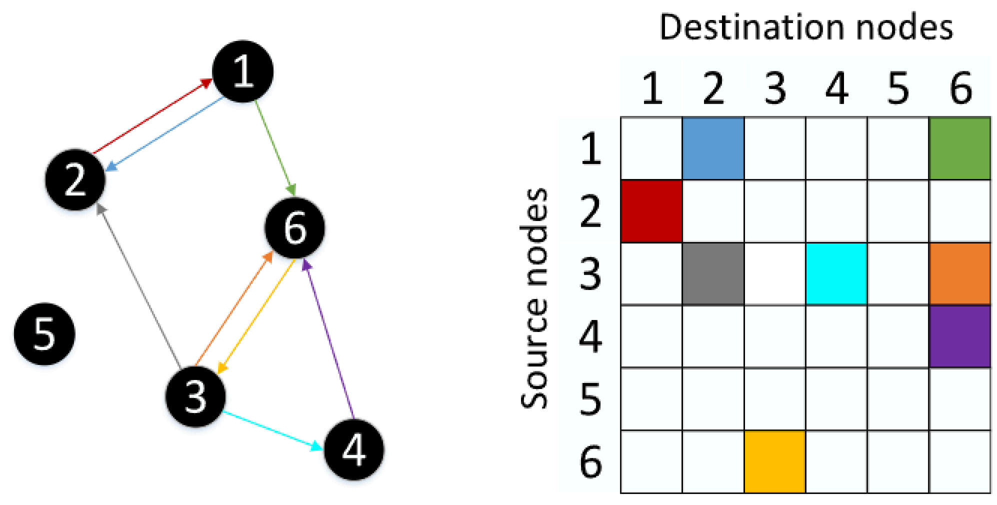

Fig. 1. Example of a directed graph and the corresponding adjacency matrix, where the left part is the connections of six nodes and right part is the adjacency matrix.

To alleviate the limitations of the GCNN method, in this paper we propose a novel dynamical graph convolutional neural networks (DGCNN) model for learning discriminative EEG features as well as the intrinsic relationship, e.g., the functional relationship, among the various EEG channels. Specifically, to learn the relationships among the various EEG channels, we propose a novel method to construct the connections among the various vertex nodes of the graph by learning an adjacency matrix. However, different from the traditional GCNN method that predetermines the adjacency matrix before the model training, the proposed DGCNN method learns the adjacency matrix in a dynamic way, i.e., the entries of the adjacency matrix are adaptively updated with the changes of graph model parameters during the model training. Consequently, in contrast to the GCNN method, the adjacency matrix learned by the DGCNN would be more useful because it captures the intrinsic connections of the EEG channels and hence it would be able to improve the discriminant abilities of the networks.

The remainder of this paper is organized as follows: In Section 2, we briefly review the preliminaries of graph theory. In Section 3, we propose the DGCNN model and the EEG emotion recognition method based on this model. Extensive experiments are conducted in Section 4. Finally, we conclude the paper in Section 5.

2 Graph Preliminary

In this section, we introduce some preliminary knowledge about the graph representation and the spectral graph filtering, which are the basis for our DGCNN method.

2.1 Graph Representation

A directed and connected graph can be defined as $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, W\}$, in which $\mathcal{V}$ represents the set of nodes with the number of $|\mathcal{V}| = N$ and $\mathcal{E}$ denotes the set of edges connecting these nodes. Let $\mathbf{W} \in \mathbb{R}^{N\times N}$ denote an adjacency matrix describing the connections between any two nodes in $\mathcal{V}$, in which the entry of $\mathbf{W}$ in the ith row and jth column, denoted by w_{ij}, measures the importance of the connection between the ith node and the jth one. Fig. 1 illustrates an example of a graph containing six vertex nodes and the

从频域角度捕获脑电图情绪信息。最常用的频域特征提取方法之一是将脑电图信号分解成多个频段，例如 d 频段（1-3 Hz）、u 频段（4-7 Hz）、a 频段（8-13 Hz）、b 频段（14-30 Hz）和 g 频段（31-50 Hz）[21], [26], [27], [28]，然后分别从每个频段提取脑电图特征。常用的脑电图特征包括差分熵（DE）特征[29], [30]、功率谱密度（PSD）特征[31]、差分不对称（DASM）特征[24]、合理不对称（RASM）特征[32]和差分尾性（DCAU）特征[19]。

为了处理脑电图情绪分类问题，文献[33]中出现了许多方法，其中使用深度神经网络（DNN）的方法[19]已被证明是最成功的方法之一。卷积神经网络（CNN）是最著名的深度神经网络方法之一，已被广泛用于处理各种分类问题，如图像分类[34]、[35]、[36]、[37]、目标检测[38]、跟踪[39]和分割[40]。尽管 CNN 模型在处理分类问题方面已被证明非常强大，但值得注意的是，以往 CNN 的应用更侧重于从图像、视频和语音中学习局部特征，其中信号的数据点会连续变化。对于其他特征学习问题，例如从交通网络和脑网络中学习特征，传统的 CNN 方法可能不太适用，因为信号在空间域中是离散和间断的。在这种情况下，基于图的方法[41]、[42]将提供更合适的方式。

图神经网络（GNN）[43]旨在基于图论构建神经网络，以处理图域中的数据。图卷积神经网络（GCNN）[44]是传统 CNN 方法的一种扩展，通过结合 CNN 与谱论[45]实现。与经典 CNN 方法相比，GCNN 在处理离散空间域中信号的特征提取方面更具优势[46]。更重要的是，GCNN 方法提供了一种有效描述图中不同节点之间内在关系的方式，这为探索脑电图（EEG）情绪识别过程中多个 EEG 通道之间的关系提供了潜在途径。

受 GCNN 模型成功的启发，本文将采用图表示方法研究多通道脑电图情绪识别问题，其中每个脑电图通道对应一个顶点节点，而两个不同顶点节点之间的连接对应图的边。尽管 GCNN 可以根据节点的空间位置描述不同节点之间的连接，但在将其应用于构建情绪识别模型之前，我们需要预先确定各种脑电图通道之间的连接。另一方面，值得注意的是，脑电图通道之间的空间位置连接与它们之间的功能连接是不同的。换句话说，更近的空间关系并不一定保证更近的功能关系。

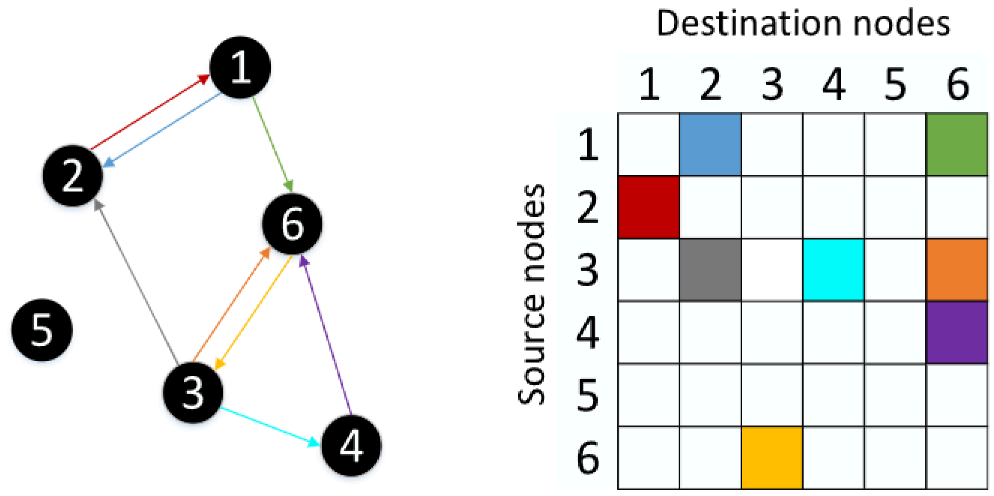

图 1. 有向图的示例及其对应的邻接矩阵，左侧是六个节点的连接，右侧是邻接矩阵。

而功能关系对于情绪识别中的判别性脑电图特征提取是有用的。因此，根据图节点的空间位置预先确定其连接是不合理的。

为了缓解 GCNN 方法的局限性，本文提出了一种新的动态图卷积神经网络（DGCNN）模型，用于学习具有判别性的脑电图（EEG）特征以及各种 EEG 通道之间的内在关系，例如功能关系。具体来说，为了学习各种 EEG 通道之间的关系，我们提出了一种通过学习邻接矩阵来构建图中各个顶点节点之间连接的新方法。然而，与传统的 GCNN 方法在模型训练前预先确定邻接矩阵不同，所提出的 DGCNN 方法以动态方式学习邻接矩阵，即邻接矩阵的元素会随着模型训练过程中图模型参数的变化而自适应更新。因此，与 GCNN 方法相比，DGCNN 学习到的邻接矩阵将更有用，因为它捕捉了 EEG 通道的内在连接，从而能够提高网络的判别能力。

本文的其余部分安排如下：在第二节中，我们简要回顾了图论的基础知识。在第三节中，我们提出了 DGCNN 模型以及基于该模型的大脑电图情绪识别方法。在第四节中，我们进行了大量的实验。最后，在第五节中，我们对全文进行总结。

2 图论基础

在本节中，我们介绍了关于图表示和谱图滤波的一些基础知识，这些是 DGCNN 方法的基础。

2.1 图表示

一个有向连通图可以定义为 $G = \{V, E, W\}$，其中 V 表示节点集合，节点数量为 $|V| = N$，E 表示连接这些节点的边集合。设 $W \in R$ 表示描述 V 中任意两个节点之间连接的邻接矩阵，其中 W 在第 i 行第 j 列的元素 w 表示第 i 个节点与第 j 个节点之间连接的重要性。图 1 展示了一个包含六个顶点节点的图示例和

edges connecting the nodes of the graph, as well as the adjacency matrix associated with the graph, where the different color arrows in the left-hand side of the figure denote the edges connecting the source nodes to destination nodes, whereas the right-hand side of the figure is the illustration of the corresponding adjacency matrix.

The commonly used methods to determine the entries w_{ij} of the adjacency matrix $\mathbf{W}$ include the distance function method [47] and K-nearest neighbor (KNN) rule method [48]. A typical distance function would be the Gaussian kernel function, which can be expressed as

$$w_{ij} = \begin{cases} \exp(-\frac{[\mathrm{dist}(i,j)]^2}{2\theta^2}), & \text{if } \mathrm{dist}(i,j) \le \tau \\ 0, & \text{otherwise} \end{cases}, \tag{1}$$

where τ and θ are two fixed parameters and $\mathrm{dist}(i,j)$ denotes the distance between the ith node and the jth one.

2.2 Spectral Graph Filtering

The spectral graph theory has been successfully used for building expander graphs [49], spectral clustering [50], graph visualization [51] and other applications [52]. Spectral graph filtering, also called graph convolution, is a popular signal processing method for graph data operation, in which Graph Fourier Transform (GFT) [47] is a typical example.

Let $\mathbf{L}$ denote the Laplacian matrix of the graph $\mathcal{G}$. Then, $\mathbf{L}$ can be expressed as

$$\mathbf{L} = \mathbf{D} - \mathbf{W} \in \mathbb{R}^{N\times N}, \tag{2}$$

where $\mathbf{D} \in \mathbb{R}^{N\times N}$ is a diagonal matrix and the ith diagonal element can be calculated by $D_{ii} = \sum_j w_{ij}$.

For a given spatial signal $\mathbf{x} \in \mathbb{R}^N$, its GFT is expressed as follows:

$$\hat{\mathbf{x}} = \mathbf{U}^T\mathbf{x}, \tag{3}$$

where $\hat{\mathbf{x}}$ denotes the transformed signal in the frequency domain, $\mathbf{U}$ is an orthonormal matrix that can be obtained via the singular value decomposition (SVD) of the graph Laplacian matrix $\mathbf{L}$ [45]

$$\mathbf{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T, \tag{4}$$

in which the columns of $\mathbf{U} = [\mathbf{u}_0, \ldots, \mathbf{u}_{N-1}] \in \mathbb{R}^{N\times N}$ constitute the Fourier basis, and $\mathbf{\Lambda} = \mathrm{diag}([\lambda_0, \ldots, \lambda_{N-1}])$ is a diagonal matrix.

From (3), the inverse of GFT can be expressed as the following form:

$$\mathbf{x} = \mathbf{U}\mathbf{U}^T\mathbf{x} = \mathbf{U}\hat{\mathbf{x}}. \tag{5}$$

Then, the convolution of two signals $\mathbf{x}$ and $\mathbf{y}$ on the graph, denoted by $\mathbf{x} *_{\mathcal{G}} \mathbf{y}$, can be expressed as [44]

$$\mathbf{x} *_{\mathcal{G}} \mathbf{y} = \mathbf{U}((\mathbf{U}^T\mathbf{x}) \odot (\mathbf{U}^T\mathbf{y})), \tag{6}$$

where $\odot$ denotes the element-wise Hadamard product.

Now let $g(\cdot)$ be a filtering function and then a signal $\mathbf{x}$ filtered by $g(\mathbf{L})$ can be expressed as

$$\mathbf{y} = g(\mathbf{L})\mathbf{x} = g(\mathbf{U}\mathbf{\Lambda}\mathbf{U}^T)\mathbf{x} = \mathbf{U}g(\mathbf{\Lambda})\mathbf{U}^T\mathbf{x}, \tag{7}$$

where $g(\mathbf{\Lambda})$ is expressed as

$$g(\mathbf{\Lambda}) = \begin{bmatrix} g(\lambda_0) & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & g(\lambda_{N-1}) \end{bmatrix}. \tag{8}$$

It is notable that the filtering operation of (7) is equivalent to the graph convolution of the signal $\mathbf{x}$ with the vector of $\mathbf{U}\mathrm{diag}(g(\mathbf{\Lambda}))$ due to the following formulation:

$$\begin{aligned} \mathbf{y} = g(\mathbf{L})\mathbf{x} &= \mathbf{U}g(\mathbf{\Lambda})\mathbf{U}^T\mathbf{x} \\ &= \mathbf{U}(\mathrm{diag}(g(\mathbf{\Lambda})) \odot (\mathbf{U}^T\mathbf{x})) \\ &= \mathbf{U}((\mathbf{U}^T(\mathbf{U}\mathrm{diag}(g(\mathbf{\Lambda}))) \odot (\mathbf{U}^T\mathbf{x})) \\ &= \mathbf{x} *_{\mathcal{G}} (\mathbf{U}\mathrm{diag}(g(\mathbf{\Lambda}))). \end{aligned} \tag{9}$$

3 DGCNN for EEG Emotion Recognition

In this section, we first propose the DGCNN model and then apply it to the EEG emotion recognition problem, in which the adjacency matrix $\mathbf{W}$ that characterizes the relationships of the various vertex nodes is dynamically learned instead of being predetermined [44].

3.1 DGCNN Model for EEG Emotion Recognition

Let $\mathbf{W}^*$ denote the optimal adjacency matrix to be learned. Then, the graph convolution of the signal $\mathbf{x}$ with the vector of $\mathbf{U}^*\mathrm{diag}(g(\mathbf{\Lambda}^*))$ defined by the spatial filtering $g(\mathbf{L}^*)$ can be expressed as

$$\mathbf{y} = g(\mathbf{L}^*)\mathbf{x} = \mathbf{U}^*g(\mathbf{\Lambda}^*)\mathbf{U}^{*T}\mathbf{x}, \tag{10}$$

where the $\mathbf{L}^*$ can be calculated from $\mathbf{W}^*$ based on (2), and $\mathbf{\Lambda}^* = \mathrm{diag}([\lambda_0^*, \ldots, \lambda_{N-1}^*])$ is a diagonal matrix.

Since it is difficult to directly calculate the expression of $g(\mathbf{\Lambda}^*)$, we simplify this calculation by using the K order Chebyshev polynomials [44] to replace the polynomial expansion of $g(\mathbf{\Lambda}^*)$, such that the calculation becomes much easier and faster. Specifically, let λ_{max}^* denote the largest element among the diagonal entries of $\mathbf{\Lambda}^*$ and $\tilde{\mathbf{\Lambda}}^*$ denote the normalized $\mathbf{\Lambda}^*$, i.e., $\tilde{\mathbf{\Lambda}}^* = 2\mathbf{\Lambda}^*/\lambda_{max}^* - \mathbf{I}_N$, such that the diagonal elements of $\tilde{\mathbf{\Lambda}}^*$ lie in the interval of $[-1, 1]$, where $\mathbf{I}_N$ is the $N \times N$ identity matrix.

Under the K order Chebyshev polynomials framework, we obtain that $g(\mathbf{\Lambda}^*)$ can be approximated by

$$g(\mathbf{\Lambda}^*) = \sum_{k=0}^{K-1} \theta_k T_k(\tilde{\mathbf{\Lambda}}^*), \tag{11}$$

where θ_k is the coefficient of Chebyshev polynomials, and $T_k(x)$ can be recursively calculated according to the following recursive expressions:

$$\begin{cases} T_0(x) = 1, \ T_1(x) = x, \\ T_k(x) = 2xT_{k-1}(x) - T_{k-2}(x), \quad k \ge 2. \end{cases} \tag{12}$$

According to (11), we obtain that the graph convolution operation defined in (10) can be rewritten by

连接图中节点的边，以及与图相关的邻接矩阵，其中左侧图中不同颜色的箭头表示连接源节点到目标节点的边，而右侧图是相应邻接矩阵的示意图。

确定邻接矩阵 W 的元素 w 的常用方法包括距离函数法[47]和 K 近邻（KNN）规则法[48]。典型的距离函数是高斯核函数，可以表示为

w¼ (expð dist½ 2u2 Š Þ; 如果 distði; jÞ t ; 0; 否则 (1)

其中 t 和 u 是两个固定参数，distði; jÞ 表示第 i 个节点和第 j 个节点之间的距离。

2.2 图谱图滤波

谱图论已成功用于构建扩展图[49]、谱聚类[50]、图可视化[51]及其他应用[52]。谱图滤波，也称为图卷积，是一种用于图数据操作的流行信号处理方法，其中图傅里叶变换（GFT）[47]是一个典型例子。

设 L 为图 G 的拉普拉斯矩阵。那么，L 可以表示为

L ¼ D W 2 R; (2)

其中 D 2 Ris 一个对角矩阵，其第 i 个对角元素可以通过 D¼ P j w 计算。

对于一个给定的空间信号 $x \in R$，其 GFT 表示如下：

x̂ = Ux; (3)

其中^x 表示频域中的变换信号，U 是一个正交矩阵，可以通过图拉普拉斯矩阵 L 的奇异值分解（SVD）获得[45]

L = ULLU; (4)

其中 U 的列向量½u; . . . ; uŠ 2 R 构成傅里叶基，而 LL ¼ diagð½; . . . ; ŠÞ是一个对角矩阵。

由(3)式，GFT 的逆可以表示为以下形式：

x ¼ UUx ¼ U^x: (5)

然后，图上两个信号 x 和 y 的卷积，记作 x y，可以表示为[44]

x y = U(Ux) (Uy); (6)

其中·表示逐元素 Hadamard 乘积。

现在设 g(·)为一个滤波函数，那么信号 x 经过 g(L)滤波后可以表示为

y = g(L)x = g(ULLU)x = Ug(LL)Ux; (7)

gðLLÞ 表示为

gðLLÞ ¼ 2 6 4 gð÷Þ 0 : . . : 0 gð÷ Þ 3 7 5: (8)

值得注意的是，公式(7)的滤波操作等同于信号 x 与 Udiag(g(L,L))向量的图卷积，原因如下：

y = g(L)x = Ug(L)Lx
= U(diag(g(L,L)))(Ux)
= U((U(U(diag(g(L,L)))(Ux))))
= x (U(diag(g(L,L)))); (9)

3 DGCNN 用于脑电图情绪识别

在本节中，我们首先提出了 DGCNN 模型，然后将其应用于脑电图情绪识别问题，其中表征各种顶点节点之间关系的邻接矩阵 W 是动态学习的，而不是预先设定的[44]。

3.1 用于脑电图情绪识别的 DGCNN 模型

设 W 为需要学习的最优邻接矩阵。那么，信号 x 与由空间滤波器 g(L)定义的向量 Udiag{g(L)}的图卷积可以表示为

y = g(L)x = Ug(L)LUx; (10)

在(2)的基础上，L 可以从 W 计算得出，

L¼ diagð½ ¥ ; . . . ; ¥ŠÞ是一个对角矩阵。

由于直接计算 gðLLÞ的表达式比较困难，我们通过使用 K 阶切比雪夫多项式[44]来替代 gðLLÞ的多项式展开，从而简化这一计算，使得计算变得更容易和更快。具体来说，设¥表示 LL 对角线元素中的最大值，~LL 表示归一化的 LL，即~LL¼ 2LL=¥I，使得~LL 的对角元素位于区间½¥1; 1Š，其中 I 是 N N 的单位矩阵。

在 K 阶切比雪夫多项式框架下，我们得到 gðLLÞ可以近似为

gðxÞ ¼ X K1 k¼0 uTðxÞ; (11)

其中 u 是切比雪夫多项式的系数，TðxÞ 可以根据以下递归公式递归计算：

TðxÞ ¼ 1; TðxÞ ¼ x; T ðxÞ ¼ 2xTðxÞ TðxÞ; k 2: (12)

根据(11)，我们得到定义在(10)中的图卷积操作可以重写为

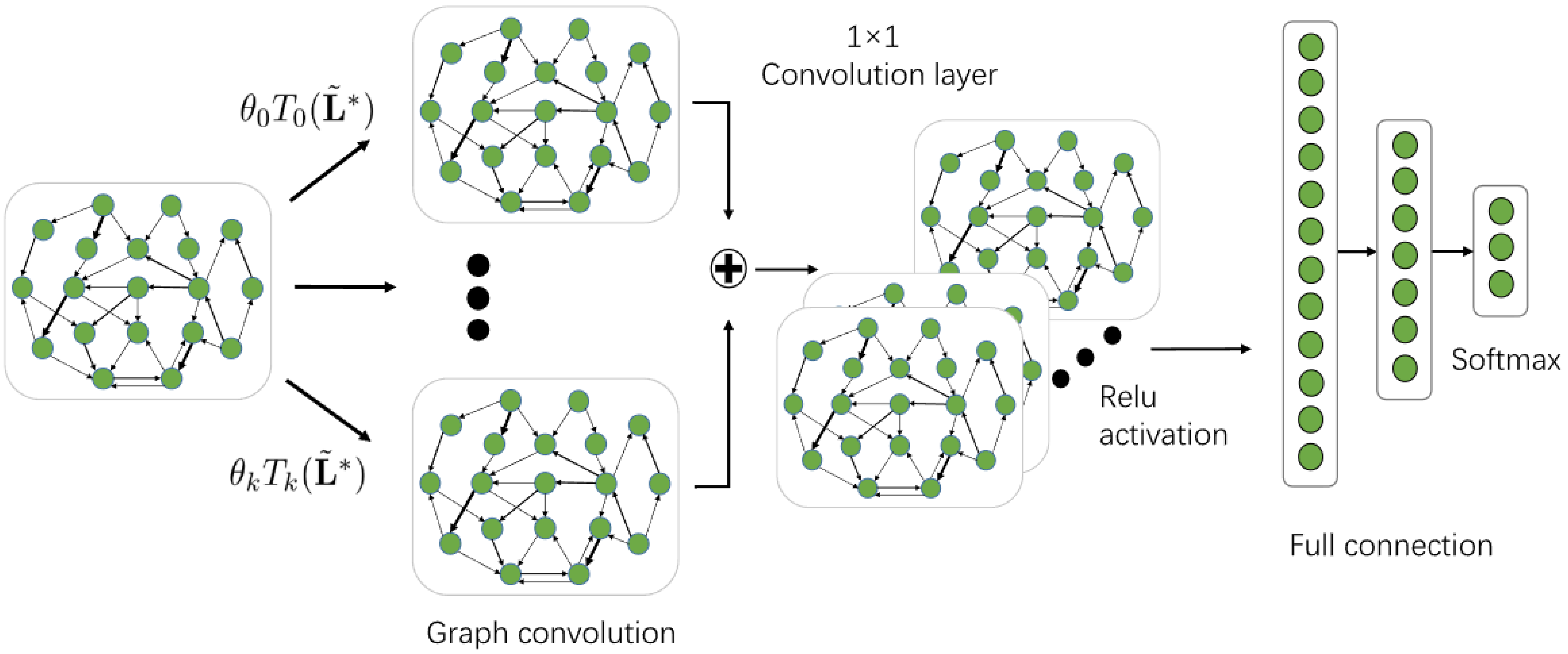

Fig. 2. The framework of the DGCNN model for EEG emotion recognition, which consists of the dynamical graph convolutional operation via the learned graph connections, convolution layer with 1×1 kernel, Relu activation and the full connection. The inputs of the model are the EEG features extracted from multiple frequency bands, e.g., five frequency bands (δ band, θ band, α band, β band, and γ band), in which each EEG channel is represented as a node of the graph. The outputs are the predicted labels through softmax.

$$\begin{aligned}\mathbf{y} &= \mathbf{U}^* g(\boldsymbol{\Lambda}^*)\mathbf{U}^{*T}\mathbf{x} \\ &= \sum_{k=0}^{K-1} \mathbf{U}^* \begin{bmatrix} \theta_k T_k(\tilde{\lambda}_0^*) & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \theta_k T_k(\tilde{\lambda}_{N-1}^*) \end{bmatrix} \mathbf{U}^{*T}\mathbf{x} \\ &= \sum_{k=0}^{K-1} \theta_k T_k(\tilde{\mathbf{L}}^*)\mathbf{x},\end{aligned} \tag{13}$$

where $\tilde{\mathbf{L}}^* = 2\mathbf{L}^*/\lambda_{max}^* - \mathbf{I}_N$.

The expression of (13) means that calculating the graph convolution of $\mathbf{x}$ can be expressed as the combination of the convolutional results of $\mathbf{x}$ with each of the Chebyshev polynomial components. Based on the expression of (13), we propose the DGCNN model for EEG emotion recognition. The framework of the proposed DGCNN model is illustrated in Fig. 2, which consists of four major layers, i.e., the dynamical graph convolutional layer, the 1×1 convolutional layer, the Relu activation layer, and the full connection layer.

Specifically, the input of the DGCNN model corresponds to the EEG features extracted from multiple frequency bands, e.g., five frequency bands (δ band, θ band, α band, β band, and γ band), in which each EEG channel is represented as a node in the DGCNN model. Following the graph filtering operation is a 1×1 convolution layer, which aims to learn the discriminative features among the various frequency domains. Moreover, to realize the nonlinear mapping capability of the network, the Relu activation function [53] is adopted to ensure that the outputs of the graph filtering layer are non-negative. Finally, the outputs of activation function are further sent to a multi-layer full connection network and a softmax function is also used to predict the desired class label information of the input EEG features.

3.2 Algorithm for DGCNN

To optimize the optimal network parameters, we adopt the back propagation (BP) method to iteratively update the network parameters until the optimal or suboptimal solutions are achieved. For this purpose, we define a loss function based on cross entropy cost, which is expressed as the following form:

$$Loss = cross\_entropy(\mathbf{l}, \mathbf{l^p}) + \alpha\|\boldsymbol{\Theta}\|, \tag{14}$$

where $\mathbf{l}$ and $\mathbf{l}^p$ denote the actual label vector of training data and the predicted one, respectively, $\boldsymbol{\Theta}$ denotes all the model parameters and α is the trade-off regularization weight. The cross entropy function $cross\_entropy(\mathbf{l}, \mathbf{l^p})$ aims at measuring the dissimilarity between the actual emotional labels and the desired ones while the regularization $\alpha\|\boldsymbol{\Theta}\|$ aims to prevent over-fitting of the model parameters learning.

When applying the BP method to dynamically learn the optimal adjacency matrix $\mathbf{W}^*$ of the DGCNN model, we have to calculate the partial derivative of the loss function with respect to $\mathbf{W}^*$, which is formulated as

$$\frac{\partial Loss}{\partial \mathbf{W}^*} = \begin{pmatrix} \frac{\partial Loss}{\partial w_{11}^*} & \cdots & \frac{\partial Loss}{\partial w_{1N}^*} \\ \vdots & \vdots & \vdots \\ \frac{\partial Loss}{\partial w_{N1}^*} & \cdots & \frac{\partial Loss}{\partial w_{NN}^*} \end{pmatrix}, \tag{15}$$

where w_{ij}^* denotes the ith row and j-th column element of $\mathbf{W}^*$. According to the chain rule, the calculation of $\frac{\partial Loss}{\partial w_{ij}^*}$ can be expressed as

$$\frac{\partial Loss}{\partial w_{ij}^*} = \frac{\partial cross\_entropy(\mathbf{l}, \mathbf{l^p})}{\partial \tilde{\mathbf{L}}^*} \cdot \frac{\partial \tilde{\mathbf{L}}^*}{\partial w_{ij}^*} + \alpha \frac{\partial \|\boldsymbol{\Theta}\|}{\partial w_{ij}^*}. \tag{16}$$

After calculating the partial derivative of $\frac{\partial Loss}{\partial \mathbf{W}^*}$, we can use the following rule to update the optimal adjacency matrix $\mathbf{W}^*$:

$$\mathbf{W}^* = (1-\rho)\mathbf{W}^* + \rho\frac{\partial Loss}{\partial \mathbf{W}^*},$$

where ρ denotes the learning rate of the network.

Algorithm 1 summarizes the detailed procedures of training the DGCNN model in EEG emotion recognition.

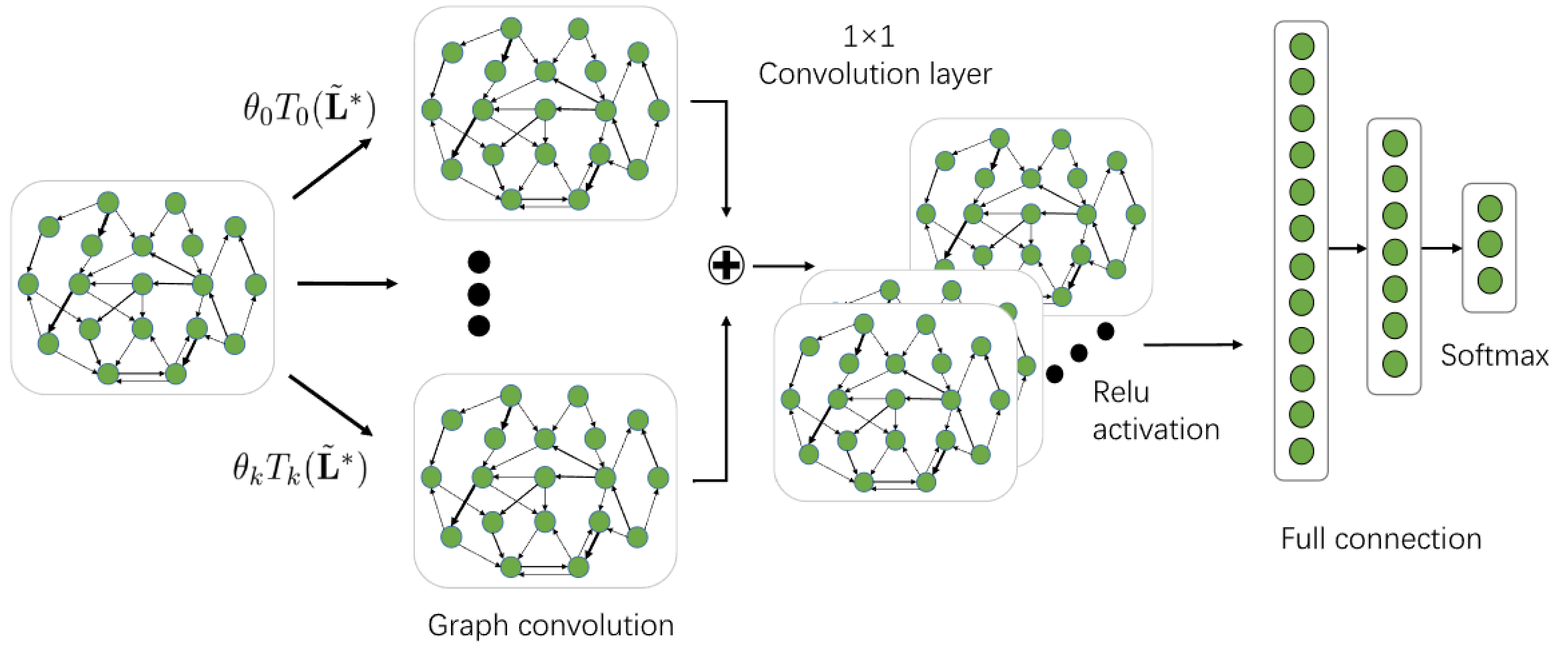

图 2. 用于脑电图情绪识别的 DGCNN 模型框架，该框架包括通过学习到的图连接进行的动态图卷积操作、1×1 卷积核的卷积层、Relu 激活和全连接。模型的输入是从多个频段提取的脑电图特征，例如五个频段（d 频段、u 频段、a 频段、b 频段和 g 频段），其中每个脑电图通道表示为图中的一个节点。输出是通过 softmax 预测的标签。

y ¼ UgðLLÞUx
2 6 6 4 3 7 7 5Ux
uTð ~ Þ 0
X K1
¼ U : ⋱ :
k¼0
0 uTð ~ Þ (13)
X K1
¼ uTð~LÞx;
k¼0

在 ~L¼ 2L= I: 式(13)的表达式意味着计算 x 的图卷积可以表示为 x 与每个切比雪夫多项式分量卷积结果的组合。基于式(13)的表达式，我们提出了用于脑电图情绪识别的 DGCNN 模型。所提出的 DGCNN 模型的框架如图 2 所示，它由四个主要层组成，即动态图卷积层、1×1 卷积层、Relu 激活层和全连接层。

具体来说，DGCNN 模型的输入对应于从多个频段提取的脑电图特征，例如五个频段（d 频段、u 频段、a 频段、b 频段和 g 频段），其中每个脑电图通道在 DGCNN 模型中表现为一个节点。图滤波操作后是一个 1×1 卷积层，旨在学习不同频域之间的判别特征。此外，为了实现网络的非线性映射能力，采用了 Relu 激活函数[53]以确保图滤波层的输出为非负值。最后，激活函数的输出进一步送入多层全连接网络，并使用 softmax 函数来预测输入脑电图特征的期望类别标签信息。

3.2 DGCNN 算法

为了优化最佳网络参数

我们采用反向传播（BP）方法迭代更新网络参数，直到达到最优或次优解。为此，我们基于交叉熵成本定义了一个损失函数，其表达式如下：

损失¼交叉熵ðl; l Þ＋akQk; (14)

其中 l 和 l 分别表示训练数据的实际标签向量和预测标签向量，Q 表示所有模型参数，a 是权衡正则化权重。交叉熵函数交叉熵ðl; lÞ旨在测量实际情感标签与期望标签之间的差异度，而正则化 akQk 旨在防止模型参数学习过拟合。

当应用 BP 方法动态学习 DGCNN 模型的最优邻接矩阵 W 时，我们必须计算损失函数相对于 W 的偏导数，其表达式为

0 B B @ @Loss @w11 @Loss @w1N 1 C C A;
@Loss : ⋱ : (15)
@W¼ @Loss @wN1 @Loss @wNN

其中 w 表示 W 的第 i 行第 j 列元素。根据链式法则，@的计算可以表示为

@Loss @cross entropyðl; l Þ @~L @wþ @akQk
@w¼ @~L @w: (16)

在计算@的偏导数后，我们可以使用以下规则来更新最优邻接矩阵 W：

W¼ ð1 rÞWþ r @Loss
@W;

其中 r 表示网络的学习率。

算法 1 总结了在脑电图情绪识别中训练 DGCNN 模型的详细步骤。

Algorithm 1. Procedures of Training Optimal DGCNN Model for EEG Emotion Recognition

Require: Multichannel EEG features associated with multiple frequency bands, the class labels corresponding to the EEG features, the number of Chebyshev polynomial order K, the learning rate ρ;
Ensure: The desired adjacency matrix $\mathbf{W}^*$ and model parameters of DGCNN;
1: Initialize the adjacency matrix $\mathbf{W}^*$ and other model parameters;
2: **repeat**
3: Regularizing the elements of the matrix $\mathbf{W}^*$ using Relu operation such that the elements are non-negative;
4: Calculating the Laplacian matrix $\mathbf{L}^*$;
5: Calculating the normalized Laplacian matrix $\tilde{\mathbf{L}}^*$;
6: Calculating the Chebyshev polynomial items $T_k(\tilde{\mathbf{L}}^*)$ $(k = 0, 1, \dots, K-1)$;
7: Calculating $\sum_{k=0}^{K-1} \theta_k T_k(\tilde{\mathbf{L}}^*)\mathbf{x}$;
8: Calculating the 1×1 convolution results and regularizing the results using the Relu operation;
9: Calculating the results of the full connection layer;
10: Calculating the loss function using (14);
11: Updating the adjacency matrix

$$\mathbf{W}^* = (1-\rho)\mathbf{W}^* + \rho \frac{\partial Loss}{\partial \mathbf{W}^*}$$

and other model parameters;
12: **until** the iterations satisfy the predefined algorithm convergence condition.

4 Experiments

In this section, we conduct extensive experiments on two emotional EEG databases that are commonly used in EEG emotion recognition to evaluate the effectiveness of the proposed GDCNN method. The first one is the SJTU Emotion EEG Database (SEED) [19] while the second one is the DREAMER [54].

4.1 Emotional EEG Databases

The SEED database contains EEG data of 15 subjects (7 males and 8 females), which are collected via 62 EEG electrodes from the subjects when they are watching fifteen Chinese film clips with three types of emotions, i.e., negative, positive and neutral. To avoid making subjects fatigue, the whole experiment will not last for a long time and the duration of each film clip is about 4 minutes. The stimulus materials can be understood without explanation. Consequently, all the EEG signals will be categorized into one of three kinds of emotion states (positive, neutral and negative). The data collection lasted for 3 different periods corresponding to 3 sessions, and each session corresponds to 15 trials of EEG data such that there are totally 45 trials of EEG data for each subject. In addition, an additional subjective self-assessment for each subject is also carried out after the subjects watching the film clips in order to guarantee that the collected EEG data share the same emotion states as the film clips presented to the subjects.

The DREAMER database consists of EEG data via 14 EEG electrodes of 23 subjects (14 males and 9 females). To build this database, 18 film clips are used for eliciting 9 different emotions, i.e., amusement, excitement, happiness, calmness, anger, disgust, fear, sadness and surprise. Each film clip lasts for a period time between 65 to 393s, which is thought to be sufficient for eliciting single emotions. The data collection begins with a neutral film clip watching to help the subjects return to the the neutral emotion state in each new trial of data collection and also to serve as the baseline signals. After watching a film clip, the self-assessment manikins (SAM) were used to acquire subjective assessments of valence, arousal and dominance. Finally, all the EEG signals are assigned with three binary states (low/high valence, low/high arousal and low/high dominance).

TABLE 1
Number of the Five Types of EEG Features Extracted from Each Frequency Band on SEED Database

Feature type	Number of EEG features per sample				
	δ	θ	α	β	γ
PSD	62	62	62	62	62
DE	62	62	62	62	62
DASM	27	27	27	27	27
RASM	27	27	27	27	27
DCAU	23	23	23	23	23

4.2 EEG Emotion Recognition Experiments on SEED Database

In this part, we conduct two kinds of experiments to evaluate the EEG emotion recognition performance using the proposed DGCNN method. The first kind of experiment is subject-dependent whereas the second one is subject-independent.

4.2.1 Subject-Dependent Experiments on SEED Database

In subject-dependent experiments, we adopt the same experimental protocol as that of [19] to evaluate the proposed DGCNN method. Specifically, for all the 15 trials of EEG data associated with one session of one subject, the first 9 trials of EEG data are used to serve as the training set and remaining 6 ones as the testing set. Then, the recognition accuracy corresponding to each period is obtained for each subject. Finally, the average classification accuracy and standard deviation over two sessions of all the 15 subjects are calculated.

We investigate five kinds of features to evaluate the proposed EEG emotion recognition method, i.e., the differential entropy feature, the power spectral density feature, the differential asymmetry feature, the rational asymmetry feature, and the differential caudality feature. The features are respectively extracted in each of the frequency bands (δ band, θ band, α band, β band, and γ band). Moreover, to extract the EEG feature, each trial of EEG signal flow is partitioned into a set of blocks, where each block contains 1s of EEG signals. In this case, we can extract five kinds of features from each block. The number of EEG features extracted from each frequency band corresponding to the various feature types are summarized in Table 1.

Based on the EEG features shown in Table 1, we have trained the DGCNN model shown in Fig. 2 for the EEG emotion recognition. In this case, each vertex node of the graph in the DGCNN model is associated with five EEG

算法 1. 用于 EEG 情绪识别的最优 DGCNN 模型训练步骤

~~需求：多通道与多个频段相关的 EEG 特征，与 EEG 特征对应的类别标~~
签，切比雪夫多项式的阶数 K，学习率 r；
保证：期望的邻接矩阵 W 和 DGCNN 模型参数；
1：初始化邻接矩阵 W 和其他模型参数；

2: 重复
3: 使用 Relu 正则化矩阵 W 的元素
一种操作，使得元素非负；4：计算拉普拉斯矩阵 L；5：计算归一化拉普拉斯矩阵~L；6：计算切比雪夫多项式项 T(~L)

(k = 0; 1; . . . ; K - 1);
7: 计算 P k¼0 uTð~LÞx;
8: 计算 1 1 卷积结果和
使用 Relu 操作正则化结果；9：计算全连接层的结果；10：使用公式(14)计算损失函数；11：更新邻接矩阵

W¼ ð1 rÞWþ r @Loss / @W

以及其他模型参数；12: 直到迭代满足预定义的算法收敛条件。

表 1 从五种类型的脑电图特征中提取的数量
SEED 数据库的每个频段

特征类型	每个样本的脑电图特征数量				
	d	u	a	b	g
功率谱密度	62	62	62	62	62
差异熵	62	62	62	62	62
DASM	27	27	27	27	27
RASM	27	27	27	27	27
DCAU	23	23	23	23	23

4 实验部分

在本节中，我们在两个常用的脑电情感识别数据集上进行了广泛的实验，以评估所提出的 GDCNN 方法的有效性。第一个是上海交通大学情感脑电数据库（SEED）[19]，第二个是 DREAMER [54]。

4.1 情感脑电数据库

SEED 数据库包含 15 名受试者的脑电数据（7 名男性，8 名女性），这些数据通过 62 个脑电电极收集，受试者在观看十五个具有三种情绪类型（负面、正面和中性）的中国电影片段时采集。为了避免受试者疲劳，整个实验不会持续很长时间，每个电影片段的时长约为 4 分钟。刺激材料无需解释即可理解。因此，所有脑电信号将被分类为三种情绪状态之一（正面、中性和负面）。数据采集持续了 3 个不同的时间段，对应 3 个实验场次，每个场次对应 15 次脑电数据，因此每个受试者共有 45 次脑电数据。此外，在受试者观看电影片段后，还会进行额外的主观自我评估，以确保收集到的脑电数据与呈现给受试者的电影片段具有相同的情绪状态。

DREAMER 数据库包含 23 名受试者（14 名男性，9 名女性）通过 14 个脑电电极采集的脑电数据。为了构建这个数据库 18 个视频片段用于引发 9 种不同的情绪，即愉悦、兴奋、快乐、平静、愤怒、厌恶、恐惧、悲伤和惊讶。每个视频片段的时长为 65 到 393 秒，被认为足以引发单一情绪。数据收集从一个中性视频片段开始，帮助受试者在每次数据收集的新试验中恢复到中性情绪状态，同时也作为基线信号。观看完一个视频片段后，使用自我评估小人偶（SAM）来获取效价、唤醒和支配的主观评估。最后，所有脑电图（EEG）信号被分配为三种二进制状态（低/高效价、低/高唤醒和低/高支配）。

4.2 SEED 数据库上的 EEG 情绪识别实验

在这一部分，我们进行两种类型的实验来评估使用所提出的 DGCNN 方法进行 EEG 情绪识别的性能。第一种实验是受试者相关的，而第二种实验是受试者无关的。

4.2.1 SEED 数据库上的受试者相关实验

在主题相关实验中，我们采用与[19]相同的实验协议来评估所提出的 DGCNN 方法。具体来说，对于与一个受试者一个会话相关的所有 15 次 EEG 数据试验，前 9 次 EEG 数据被用作训练集，剩余的 6 次作为测试集。然后，为每个受试者获得每个时期的识别准确率。最后，计算所有 15 个受试者两个会话的平均分类准确率和标准差。

我们研究了五种特征来评估所提出的脑电图情绪识别方法，即微分熵特征、功率谱密度特征、微分不对称特征、理性不对称特征和微分尾特征。这些特征分别在每个频段（d 频段、u 频段、a 频段、b 频段和 g 频段）中提取。此外，为了提取脑电图特征，每个脑电信号流试验被划分为一组块，每个块包含 1 秒的脑电信号。在这种情况下，我们可以从每个块中提取五种特征。对应于各种特征类型的每个频段提取的脑电图特征数量汇总在表 1 中。

根据表 1 中所示的脑电图特征，我们已训练出图 2 所示的 DGCNN 模型用于脑电图情绪识别。在这种情况下，DGCNN 模型中的每个图顶点节点都与五个脑电图相关联

TABLE 2
Comparisons of the Average Accuracies and Standard Deviations (%) of Subject Dependent EEG-Based Emotion Recognition Experiments on SEED Database Among the Various Methods

Feature	Method	δ band	θ band	α band	β band	γ band	all ($\delta, \theta, \alpha, \beta, \gamma$)
DE	DBN [19]	64.32 / 12.45	60.77 / 10.42	64.01 / 15.97	78.92 / 12.48	79.19 / 14.58	86.08 / 8.34
	SVM [19]	60.50 / 14.14	60.95 / 10.20	66.64 / 14.41	80.76 /11.56	79.56 / 11.38	83.99 / 9.72
	GCNN	72.75 / 10.85	**74.40** / **8.23**	73.46 / 12.17	83.24 / 9.93	83.36 / 9.43	87.40 /9.20
	DGCNN	**74.25** / **11.42**	71.52 / 5.99	**74.43** / **12.16**	**83.65** / **10.17**	**85.73** / **10.64**	**90.40** / **8.49**
PSD	DBN [19]	60.05 / 16.66	55.03 / 13.88	52.79 / 15.38	60.68 / 21.31	63.42 / 19.66	61.90 / 16.65
	SVM [19]	58.03 / 15.39	57.26 / 15.09	59.04 / 15.75	73.34 /15.20	71.24 / 16.38	59.60 / 15.93
	GCNN	69.89 / 13.83	70.92 / 9.18	73.18 / 12.74	76.21 / 10.76	76.15 / 10.09	81.31 / 11.26
	DGCNN	**71.23** / **11.42**	**71.20** / **8.99**	**73.45** / **12.25**	**77.45** / **10.81**	**76.60** / **11.83**	**81.73** / **9.94**
DASM	DBN [19]	48.79 / 9.62	51.59 / 13.98	54.03 / 17.05	69.51 / 15.22	70.06 / 18.14	72.73 / 15.93
	SVM [19]	48.87 / 10.49	53.02 / 12.76	59.81 / 14.67	**75.03** / **15.72**	**73.59**/**16.57**	72.81 / 16.57
	GCNN	**57.07** / **6.75**	54.80 / 9.09	62.97 / 13.43	74.97 / 13.40	73.28 / 13.67	76.00 / 13.32
	DGCNN	55.93 / 9.14	**56.12** / **7.86**	**64.27** / **12.72**	73.61 / 14.35	73.50 / 16.6	**78.45** / **11.84**
RASM	DBN [19]	48.05 / 10.37	50.62 / 14.02	56.15 / 15.28	70.31 / 15.62	68.22 / 18.09	71.30 / 16.16
	SVM [19]	47.75 / 10.59	51.40 / 12.53	60.71 / 14.57	74.59 /16.18	74.61 / 15.57	74.74 / 14.79
	GCNN	**59.70** / **5.65**	**55.91** / **8.82**	59.97 / 14.27	**79.45** / **13.32**	79.73 / 13.22	84.06 / 12.86
	DGCNN	57.79 / 6.90	55.79 / 8.10	**61.58** / **12.63**	75.79 / 13.07	**82.32** / **11.54**	**85.00** / **12.47**
DCAU	DBN [19]	54.58 / 12.81	56.94 / 12.54	57.62 / 13.58	70.70 / 16.33	72.27 / 16.12	77.20 / 14.24
	SVM [19]	55.92 / 14.62	57.16 / 10.77	61.37 / 15.97	75.17 / 15.58	76.44 / 15.41	77.38 / 11.98
	GCNN	62.60 / 12.88	**65.05** / **8.35**	66.41 / 11.06	77.28 / 11.55	78.68 / 13.00	79.02 / 11.27
	DGCNN	**63.18** / **13.48**	62.55 / 7.96	**67.71** / **10.74**	**78.68** / **10.81**	**80.05** / **13.03**	**81.91** / **10.06**

features corresponding to the five frequency bands. It is notable that the number of vertex nodes in the graph would be different for different feature types. Specifically, for both PSD and DE features, the number of vertex nodes in the graph is 62 whereas it would be 27 nodes for both DASM and RASM because there are only 27 features for these two feature types. Similarly, the number of vertex nodes in the graph would be 23 for DCAU feature type.

Table 2 summarizes experimental results in terms of the average EEG emotion recognition accuracy and the standard deviation of the DGCNN method under the five different EEG feature types (PSD, DE, DASM, RASM and DCAU) and the five different frequency bands, where the results annotated by "all" mean that the EEG features associated with all of the five frequency bands are combined together in the experiments such that each vertex node of the graph is associated with five EEG features. For the comparison purpose, we also include the experimental results of [19] with deep belief networks (DBN) [55] and support vector machine (SVM) [56] in Table 2. Moreover, we also conduct the same experiments using the GCNN method to serve as a baseline method to evaluate the performance of DGCNN. Here it should be noted that, for the GCNN model, the elements of the adjacency matrix are predetermined according to the spatial relationship of the EEG channels (Fig. 3 illustrates the spatial relationship of the 62 EEG channels used to construct the adjacency matrix for the GCNN model, i.e., there would be a direct connection between two EEG channels if they are closely related.).

From Table 2, we can observe the following major points:

- Among the five kinds of EEG features, the DE feature was demonstrated to be better than most of the other ones in terms of the average recognition accuracy. For each emotion recognition method, the best average recognition accuracy was achieved when all the five frequency bands were used together. Especially, the best average recognition accuracy of DE feature was as high as 90.4 percent (with standard deviation of 8.49 percent) when all the five frequency bands were used.
- Among the four EEG emotion recognition methods, both DGCNN and GCNN achieve better average recognition accuracies than the other two methods (SVM and DBN). Compared with GCNN, however, DGCNN demonstrates to be more powerful in classifying the EEG emotion classes in most cases when the same EEG feature is used. This is most likely due to the optimization for determining the entries of the adjacency matrix in the GCNN model, which makes

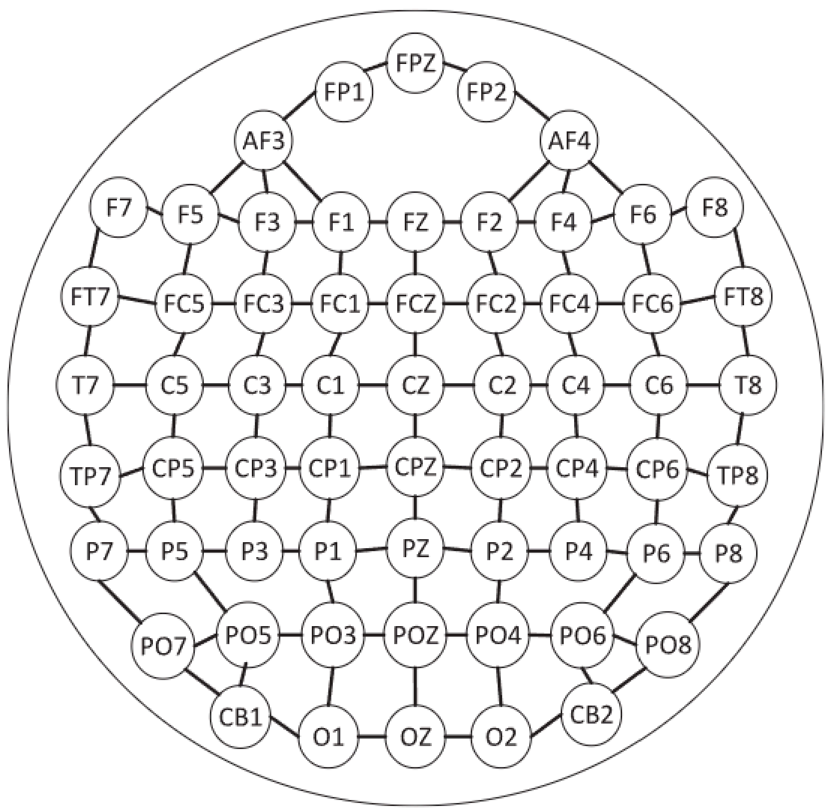

Fig. 3. Illustration of the connections among the 62 EEG channels, which is used for constructing the adjacency matrix of GCNN.

表 2 在 SEED 数据库上各种方法中，基于受试者依赖的脑电图情绪识别实验的平均准确率和标准偏差（%）的比较

特征 方法 d 波段 u 波段 a 波段 b 波段 g 波段 全部(d; u; a; b; g) DE	DBN [19] 64.32 / 12.45 60.77 / 10.42 64.01 / 15.97 78.92 / 12.48 79.19 / 14.58 86.08 / 8.34 SVM [19] 60.50 / 14.14 60.95 / 10.20 66.64 / 14.41 80.76 /11.56 79.56 / 11.38 83.99 / 9.72 GCNN 72.75 / 10.85 74:40 = 8:23 73.46 / 12.17 83.24 / 9.93 83.36 / 9.43 87.40 /9.20
	DGCNN 74:25 = 11:42 71.52 / 5.99 74:43 = 12:16 83:65 = 10:17 85:73 = 10:64 90:40 = 8:49
PSD	DBN [19] 60.05 / 16.66 55.03 / 13.88 52.79 / 15.38 60.68 / 21.31 63.42 / 19.66 61.90 / 16.65 SVM [19] 58.03 / 15.39 57.26 / 15.09 59.04 / 15.75 73.34 /15.20 71.24 / 16.38 59.60 / 15.93 GCNN 69.89 / 13.83 70.92 / 9.18 73.18 / 12.74 76.21 / 10.76 76.15 / 10.09 81.31 / 11.26 DGCNN 71:23 = 11:42 71:20 = 8:99 73:45 = 12:25 77:45 = 10:81 76:60 = 11:83 81:73 = 9:94
DASM	DBN [19] 48.79 / 9.62 51.59 / 13.98 54.03 / 17.05 69.51 / 15.22 70.06 / 18.14 72.73 / 15.93 SVM [19] 48.87 / 10.49 53.02 / 12.76 59.81 / 14.67 75:03 = 15:72 73:59=16:57 72.81 / 16.57 GCNN 57:07 = 6:75 54.80 / 9.09 62.97 / 13.43 74.97 / 13.40 73.28 / 13.67 76.00 / 13.32 DGCNN 55.93 / 9.14 56:12 = 7:86 64:27 = 12:72 73.61 / 14.35 73.50 / 16.6 78:45 = 11:84
RASM	DBN [19] 48.05 / 10.37 50.62 / 14.02 56.15 / 15.28 70.31 / 15.62 68.22 / 18.09 71.30 / 16.16 SVM [19] 47.75 / 10.59 51.40 / 12.53 60.71 / 14.57 74.59 /16.18 74.61 / 15.57 74.74 / 14.79 GCNN 59:70 = 5:65 55:91 = 8:82 59.97 / 14.27 79:45 = 13:32 79.73 / 13.22 84.06 / 12.86 DGCNN 57.79 / 6.90 55.79 / 8.10 61:58 = 12:63 75.79 / 13.07 82:32 = 11:54 85:00 = 12:47
DCAU	DBN [19] 54.58 / 12.81 56.94 / 12.54 57.62 / 13.58 70.70 / 16.33 72.27 / 16.12 77.20 / 14.24 SVM [19] 55.92 / 14.62 57.16 / 10.77 61.37 / 15.97 75.17 / 15.58 76.44 / 15.41 77.38 / 11.98 GCNN 62.60 / 12.88 65:05 = 8:35 66.41 / 11.06 77.28 / 11.55 78.68 / 13.00 79.02 / 11.27 DGCNN 63:18 = 13:48 62.55 / 7.96 67:71 = 10:74 78:68 = 10:81 80:05 = 13:03 81:91 = 10:06

对应于五个频段的特征。值得注意的是，图中顶点的数量会因不同的特征类型而异。具体来说，对于 PSD 和 DE 特征，图中的顶点节点数量为 62，而对于 DASM 和 RASM 特征，则为 27 个节点，因为这两种特征类型只有 27 个特征。类似地，对于 DCAU 特征类型，图中的顶点节点数量为 23。

表 2 总结了在不同五种脑电特征类型（PSD、DE、DASM、RASM 和 DCAU）和五种不同频段下，DGCNN 方法的平均脑电情绪识别准确率及标准差。其中标注为“all”的结果表示实验中结合了与所有五个频段相关的脑电特征，使得图中的每个顶点节点都与五个脑电特征相关联。为了比较，我们在表 2 中也包含了[19]使用深度信念网络（DBN）[55]和支持向量机（SVM）[56]的实验结果。此外，我们还使用 GCNN 方法进行了相同的实验，作为基线方法来评估 DGCNN 的性能。需要注意的是，对于 GCNN 模型，邻接矩阵的元素是根据脑电通道的空间关系预先确定的（图 3 展示了用于构建 GCNN 模型邻接矩阵的 62 个脑电通道的空间关系，即如果两个脑电通道密切相关，则它们之间会有直接连接）。

从表 2 中，我们可以观察到以下几点：在五种脑电图（EEG）特征中，差异熵（DE）特征在平均识别准确率方面表现优于其他大多数特征。对于每种情绪识别方法，当所有五个频段被一起使用。特别是，当使用所有五个频段时，DE 特征的最好平均识别准确率高达 90.4%（标准差为 8.49%）。

在四种脑电图情绪识别方法中，DGCNN 和 GCNN 的平均识别准确率均高于其他两种方法（SVM 和 DBN）。然而，与 GCNN 相比，DGCNN 在大多数情况下使用相同的脑电图特征时，表现出更强的分类能力。这很可能是因为 GCNN 模型中对邻接矩阵条目的优化，这使得

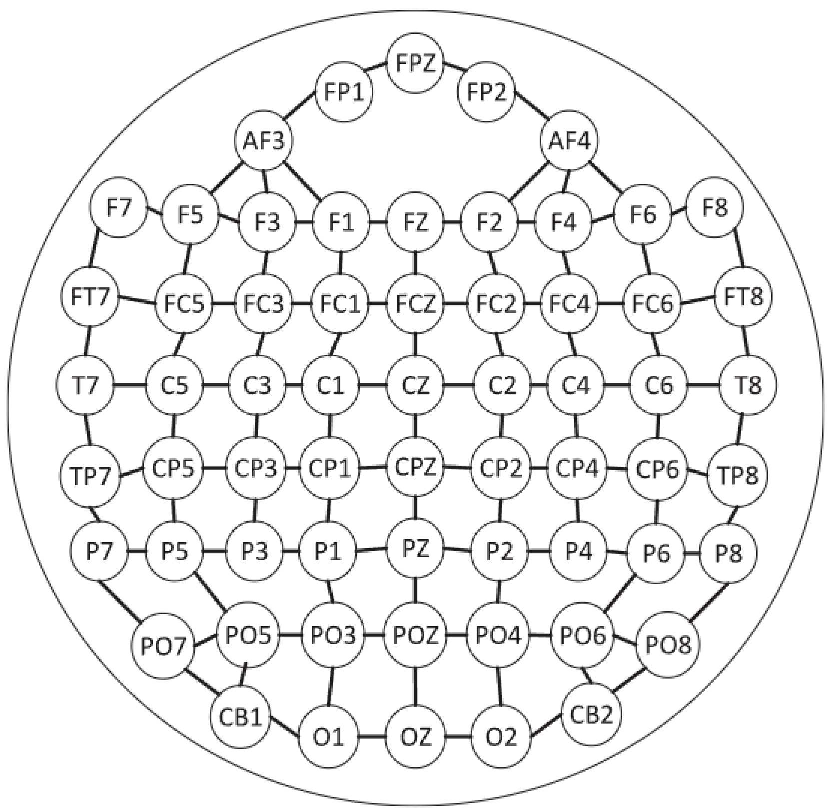

图 3. 62 个脑电图通道之间连接的示意图，用于构建 GCNN 的邻接矩阵。

TABLE 3
The Average Accuracies and Standard Deviations (%) of Subject Independent LOSO Cross Validation EEG-Based Emotion Recognition Experiments on SEED Database Using DGCNN

Method	Feature	δ band	θ band	α band	β band	γ band	all ($\delta, \theta, \alpha, \beta, \gamma$)
DGCNN	DE	49.79 / 10.94	46.36 / 12.06	**48.29 / 12.28**	56.15 /14.01	**54.87 / 17.53**	**79.95 / 9.02**
	PSD	**50.36 / 9.66**	48.85 / 9.44	43.39 / 8.22	**56.39 / 10.9**	51.81 / 10.88	64.27 / 13.80
	DASM	44.31 / 6.67	43.79 / 6.31	44.64 / 8.94	47.53 / 11.95	47.13 / 10.89	52.50 / 11.92
	RASM	43.42 / 4.07	42.69 / 7.11	42.77 / 7.43	49.52 / 10.22	46.37 / 12.1	58.46 / 10.08
	DCAU	48.52 / 10.23	**52.44 / 9.26**	44.46 / 8.39	52.55 / 13.31	50.97 / 13.66	65.19 / 10.49

it more accurate to characterize the relationship between the various EEG channels.

- Among the five EEG frequency bands, both β and γ frequency bands achieve better recognition results than the other three ones in most cases, which indicates that the higher frequency bands may be more closely related with the emotion activities whereas the lower frequency bands are less related with the emotion activities.

4.2.2 Subject-Independent Experiments on SEED

In the subject-independent experiments, we adopt the leave-one-subject-out (LOSO) cross-validation strategy to evaluate the EEG emotion recognition performance of the proposed DGCNN method. Specifically, in LOSO cross-validation experimental protocol, the EEG data of 14 subjects are used for training the model and the EEG data of the rest one subject is used as testing data. The experiments are repeated such that the EEG data of each subject are used once as the testing data. The average classification accuracies and standard deviations corresponding to the five kinds of EEG features are respectively calculated.

Table 3 summarizes the experimental results in terms of the average EEG emotion recognition accuracy and the standard deviation of DGCNN under the different kinds of EEG features and the different frequency bands. From Table 3, we obtain the following major points:

- For each kind of feature, the recognition accuracies associated with higher frequency bands are better than the ones of lower frequency bands.
- For each kind of feature, the best recognition accuracy is obtained when combining the features extracted from the five frequency bands together for emotion recognition. Especially, when the five frequency bands are used and DE feature is adopted, the best recognition accuracy ($= 79.95\%$) is achieved.

Considering that the subject-independent emotion recognition task can be seen as a cross-domain emotion recognition problem, we adopt several popular cross-domain recognition methods, including transfer component analysis (TCA) [57], KPCA [58], transductive SVM (T-SVM) [59] and transductive parameter transfer (TPT) [60], to serve as the baseline methods. Since the use of the DE features combined with the five frequency bands (δ, θ, α, β, and γ) had been demonstrated to be the most effective features in the EEG emotion recognition, we adopt these features to compare the EEG emotion recognition performance among the five methods. Fig. 4 summarizes the experimental results. From the results of Fig. 4, we can see that the proposed DGCNN method achieves the highest recognition accuracy ($= 79.95\%$) among the five methods. In addition, we can also see that the standard deviation ($= 9.02\%$) of the proposed DGCNN is much lower than that of TCA, KPCA, T-SVM, and TPT, which indicates that DGCNN is much more stable compared with the other four methods.

4.3 EEG Emotion Recognition Experiments on DREAMER Database

In this part, we conducted experiments on the DREAMER database to evaluate the EEG emotion recognition performance using the proposed DGCNN method. Before the experiments, we adopted the same feature extraction method as that of [54] to extract a set of PSD features. During the feature extraction, we first cropped the EEG signals corresponding to the last 60 seconds of each film clip and then decomposed the signals into θ (4-8 Hz), α (8-13 Hz), and β (13-20 Hz) frequency bands. For each frequency band, the 60s EEG signals are further segmented into a set of 59 blocks by sliding a window with the size of 256 EEG points, in which there are half overlap between two subsequent blocks. Finally, the PSD features were calculated from the EEG signal of each block. As a result, we obtain 14 features associated with the 14 EEG channels in total from each block and we concatenate them into a 14-dimensional feature vector to represent a EEG data sample. In this way, we totally obtain 59 EEG data samples associated with each frequency band from each session, which are used for our EEG emotion recognition evaluation of the DGCNN method.

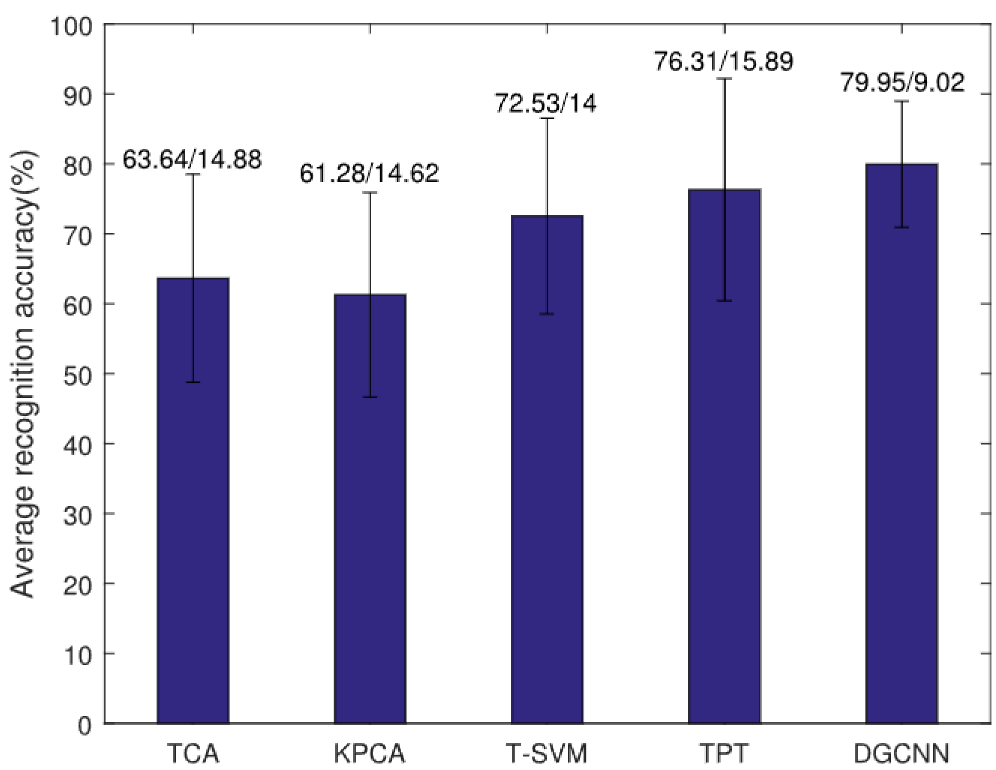

Fig. 4. Comparisons of the EEG emotion recognition accuracies and standard deviations among TCA, KPCA, T-SVM, TPT, and DGCNN on SEED database.

表 3 使用 DGCNN 在 SEED 数据库上进行的主题独立 LOSO 交叉验证脑电情感识别实验的平均准确率和标准偏差（%）

方法	特征	d 波段	U 波段	A 波段	B 波段	G 波段	所有 (d; u; a; b; g)
DGCNN	DE 49.79 / 10.94 46.36 / 12.06 48.29 / 12.28 56.15 /14.01 54.87 / 17.53 79.95 / 9.02 PSD 50.36 / 9.66 48.85 / 9.44 43.39 / 8.22 56.39 / 10.9 51.81 / 10.88 64.27 / 13.80 DASM 44.31 / 6.67 43.79 / 6.31 44.64 / 8.94 47.53 / 11.95 47.13 / 10.89 52.50 / 11.92 RASM 43.42 / 4.07 42.69 / 7.11 42.77 / 7.43 49.52 / 10.22 46.37 / 12.1 58.46 / 10.08 DCAU 48.52 / 10.23 52.44 / 9.26 44.46 / 8.39 52.55 / 13.31 50.97 / 13.66 65.19 / 10.49						

用这种方式来描述各种脑电图通道之间的关系更为准确。

在五种脑电图（EEG）频率带中，b 波和 g 波频率带在大多数情况下都取得了比其他三种频率带更好的识别结果，这表明较高频率带可能与情绪活动更密切相关，而较低频率带与情绪活动的关系较弱。

4.2.2 SEED 的受试者无关实验

在受试者无关实验中，我们采用留一受试者交叉验证（LOSO）策略来评估所提出的 DGCNN 方法的 EEG 情绪识别性能。具体来说，在 LOSO 交叉验证实验协议中，14 名受试者的 EEG 数据用于训练模型，而其余一名受试者的 EEG 数据用作测试数据。实验重复进行，使得每个受试者的 EEG 数据都作为一次测试数据使用。分别计算了与五种 EEG 特征相对应的平均分类准确率和标准差。

表 3 总结了在不同种类的脑电图（EEG）特征和不同频段下，DGCNN 的平均脑电图情绪识别准确率及其标准差。从表 3 中，我们得到以下几点主要结论：对于每种特征，识别准确率

与高频带相关的比与低频带相关的更好。

对于每种特征，当将五个频段提取的特征结合起来进行情绪识别时，可以获得最佳识别精度。特别是当使用五个频段并采用 DE 特征时，最佳识别精度（79.95%）得以实现。考虑到主体无关的情绪识别任务可以被视为跨域情绪识别问题，我们采用几种流行的跨域识别方法，包括迁移成分分析（TCA）[57]、核主成分分析（KPCA）[58]、直推支持向量机（T-SVM）[59]和直推参数迁移（TPT）[60]，作为基线方法。由于使用 DE 特征结合五个频段（d、u、a、b 和 g）已被证明是在脑电图情绪识别中最有效的特征，我们采用这些特征来比较五种方法的脑电图情绪识别性能。图 4 总结了实验结果。从图 4 的结果来看

我们可以看到，所提出的 DGCNN 方法在五种方法中实现了最高的识别准确率（¼ 79:95%）。此外，我们还可以看到，所提出的 DGCNN 的标准差（¼ 9:02%）远低于 TCA、KPCA、T-SVM 和 TPT，这表明与其他四种方法相比，DGCNN 更加稳定。

4.3 DREAMER 数据库上的脑电图情绪识别实验

在这一部分，我们使用所提出的 DGCNN 方法在 DREAMER 数据库上进行了实验，以评估脑电图（EEG）情绪识别性能。在实验之前，我们采用了与[54]相同的特征提取方法来提取一组功率谱密度（PSD）特征。在特征提取过程中，我们首先裁剪了每个电影片段最后 60 秒对应的 EEG 信号，然后将信号分解为 u（4-8 Hz）、a（8-13 Hz）和 b（13-20 Hz）三个频段。对于每个频段，60 秒的 EEG 信号通过滑动一个 256 个 EEG 点的窗口进一步分割成一组 59 个块，其中相邻块之间有 50%的重叠。最后，从每个块的 EEG 信号中计算 PSD 特征。因此，我们从每个块中获得了与 14 个 EEG 通道相关的 14 个特征，并将它们连接成一个 14 维的特征向量来表示一个 EEG 数据样本。通过这种方式，我们从每个会话中总共获得了与每个频段相关的 59 个 EEG 数据样本，用于 DGCNN 方法的 EEG 情绪识别评估。

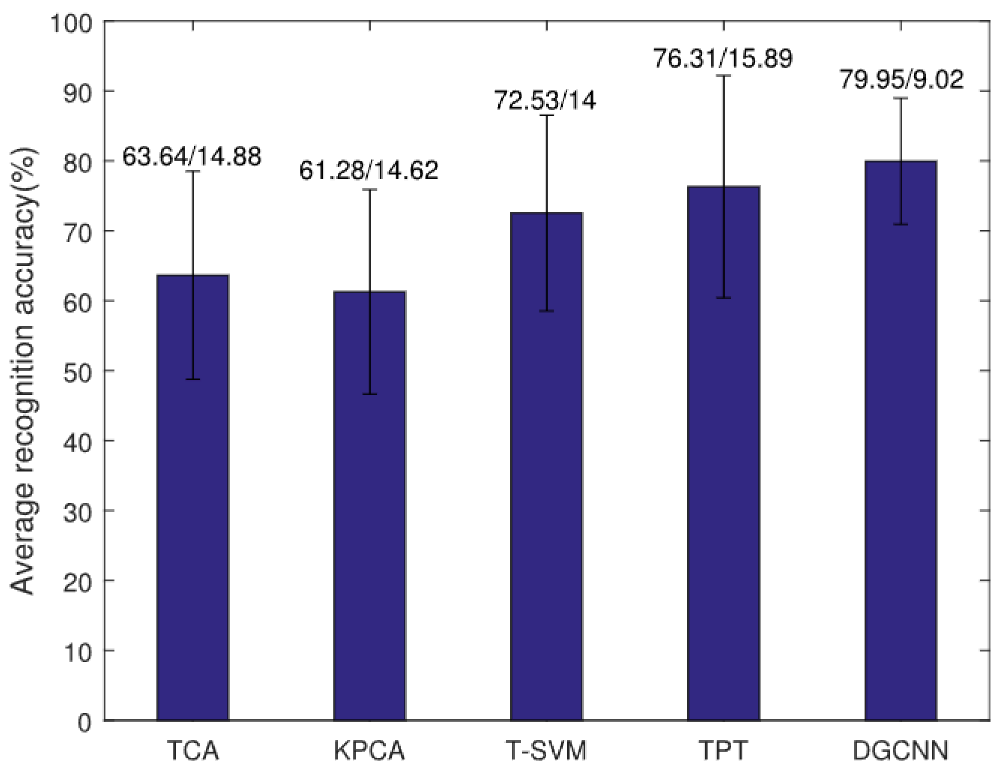

图 4. TCA、KPCA、T-SVM、TPT 和 DGCNN 在 SEED 数据库上脑电情感识别准确率和标准差的比较。

TABLE 4
Number of the PSD EEG Features of Each EEG Sample Associated with Each Frequency Band on DREAMER Database

Feature type	Number of EEG features per sample		
	θ	α	β
PSD	14	14	14

Table 4 shows the information about the EEG features we extracted from the DREAMER database.

To evaluate the EEG emotion recognition performance of the proposed DGCNN method, we adopt the subject-dependent leave-one-session-out cross-validation strategy to carry out the experiments. Specifically, for all the 18 sessions (corresponding to the 18 film clips) of experiments for each subject, we choose the EEG data belonging to one session as the testing data and use the ones belonging to the other 17 sessions as the training data. The emotion classification accuracy for the testing data is calculated based on the training data. This procedure is repeated for 18 trials such that the EEG data samples of each session have been used once as the testing data. Then, the overall classification accuracy of this subject is achieved by averaging the recognition accuracies of all the 18 trials. Finally, we use the average emotion classification accuracy obtained by averaging all the 23 subjects to evaluate the emotion recognition performance of the DGCNN method.

Table 5 shows the experimental results of the DGCNN method with respect to different emotion dimensions (i.e., valence, arousal and dominance). For comparison purpose, three state-of-the-art methods, i.e., SVM, graph regularized sparse linear discriminant analysis (GraphSLDA) [61] and group sparse canonical correlation analysis (GSCCA) [9], are also used to conduct the same experiments and the experimental results are included in Table 5.

From Table 5, we can obtain the following major points:

- The proposed DGCNN method achieves much higher classification accuracies than the other three state-of-the-art methods, in which the classification accuracies could be as high as 86.23 percent for valence classification, 84.54 percent for arousal classification, and 85.02 percent for dominance classification, respectively.
- The proposed DGCNN achieves more stable results than SVM, GraphSLDA, and GSCCA in terms of the standard deviations, in which the standard deviations of DGCNN are 12.29 percent for valence, 10.18 percent for arousal and 10.25 percent for dominance, respectively.

TABLE 5
The Average Classification Accuracies and Standard Deviations (%) of Subject-Dependent EEG Emotion Recognition on DREAMER Database Using PSD Feature

Method	Valence	Arousal	Dominance
SVM	60.14 / 33.34	68.84 / 24.94	75.84 / 20.76
GraphSLDA	57.70 / 13.89	68.12 / 17.53	73.90 / 15.85
GSCCA	56.65 / 21.50	70.30 / 18.66	77.31 / 15.44
DGCNN	**86.23 / 12.29**	**84.54 / 10.18**	**85.02 / 10.25**

5 Conclusions and Discussions

In this paper, we have proposed a novel DGCNN model for EEG emotion recognition and conducted experiments on SEED and DREAMER EEG emotion databases, respectively, to evaluate the effectiveness of the proposed method. Both subject-dependent experiments and subject-independent cross validation experiments on SEED database had been conducted and the experimental results indicated that the DGCNN method achieves better recognition performance than the state-of-the-art methods such as the SVM, DBN, KPCA, TCA, T-SVM and TPT methods. Especially, when the DE features of five frequency bands are combined together, the average recognition accuracy of DGCNN can be as high as 90.40 percent for the case of subject dependent experiments and can be as high as 79.95 percent for the case of subject independent cross validation experiments. On the DREAMER database, the average accuracies of valence, arousal, and dominance using the proposed DGCNN are 86.23, 84.54 and 85.02 percent respectively, which are higher than SVM, GraphSLDA and GSCCA. The better recognition performance of DGCNN is most likely due to the following major points:

- The use of nonlinear neural network of DGCNN makes it much more powerful to deal with the nonlinear discriminative feature learning;
- The graph representation of DGCNN provides a useful way to characterize the intrinsic relationships among the various EEG channels, which is advantageous for extracting the most discriminative features for the emotion recognition task;
- DGCNN adaptively learns the intrinsic relationships of EEG channels through optimization of the adjacency matrix **W**. In contrast to DGCNN, the GCNN model determines the values of **W** prior to the model learning stage. Consequently, using DGCNN model would be more accurate than GCNN to characterize the relationships of EEG channels.

Additionally, it is notable that the diagonal elements of the adjacency matrix indicate the contributions of the EEG channels to the EEG emotion recognition. Hence, the adjacency matrix would provide a potential way to find out what are the most contributive EEG channels in the EEG emotion recognition, which would be advantageous to further improve the EEG emotion recognition performance. We leave this interesting topic as our future work.

Another interesting issue that could be investigated in future is about the data scale generalization. Although the proposed DGCNN method had demonstrated to be a good method in dealing with the EEG emotion recognition, it is also notable that the data scales of the EEG databases used in the experiments are still relatively small, which would not be enough for learning more powerful deep neural network models and hence may limit the further performance improvement in this research. Consequently, an EEG database with much larger scales is desired in solving this problem and this is also a major task of our future work.

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表 4 DREAMER 数据库中每个脑电图样本与每个频段相关的 PSD 脑电图特征数量

特征类型	每个样本的 EEG 特征数量		
	u	a	b
PSD	14	14	14

表 4 展示了我们从 DREAMER 数据库中提取的脑电图（EEG）特征信息。

为了评估所提出的 DGCNN 方法在脑电图情绪识别方面的性能，我们采用受试者相关的留一式交叉验证策略进行实验。具体来说，对于每个受试者的所有 18 个实验会话（对应于 18 个电影片段），我们选择一个会话的脑电图数据作为测试数据，并使用其余 17 个会话的数据作为训练数据。基于训练数据计算测试数据的情绪分类准确率。该过程重复进行 18 次，使得每个会话的脑电图数据样本都作为测试数据使用一次。然后，通过平均所有 18 次试验的分类准确率来获得该受试者的总体分类准确率。最后，我们通过平均所有 23 个受试者获得的平均情绪分类准确率来评估 DGCNN 方法的情绪识别性能。

表 5 展示了 DGCNN 方法在不同情绪维度（即效价、唤醒度和支配度）上的实验结果。为了比较，还使用了三种最先进的方法，即 SVM、图正则化稀疏线性判别分析（GraphSLDA）[61]和组稀疏典型相关分析（GSCCA）[9]，进行相同的实验，实验结果也包含在表 5 中。

从表 5 中，我们可以得到以下几点主要结论：① 提出的 DGCNN 方法比其他三种最先进方法的分类准确率要高得多，其中分类准确率最高可达 86.23%（效价分类）、84.54%（唤醒度分类）和 85.02%（支配度分类）。② 在标准差方面，DGCNN 比 SVM、GraphSLDA 和 GSCCA 的结果更稳定，其中 DGCNN 的标准差分别为 12.29%（效价）、10.18%（唤醒度）和 10.25%（支配度）。

表 5 DREAMER 数据库上使用 PSD 特征进行受试者相关 EEG 情绪识别的平均分类准确率和标准差(%)

方法	效价	唤起	主导
SVM	60.14 / 33.34	68.84 / 24.94	75.84 / 20.76
GraphSLDA	57.70 / 13.89	68.12 / 17.53	73.90 / 15.85
GSCCA	56.65 / 21.50	70.30 / 18.66	77.31 / 15.44
DGCNN	86.23 / 12.29	84.54 / 10.18	85.02 / 10.25

5 结论与讨论

在这篇论文中，我们提出了一种用于脑电图情绪识别的新型 DGCNN 模型，并在 SEED 和 DREAMER 脑电图情绪数据库上分别进行了实验，以评估所提出方法的有效性。在 SEED 数据库上，我们进行了受试者相关实验和受试者无关交叉验证实验，实验结果表明 DGCNN 方法比 SVM、DBN、KPCA、TCA、T-SVM 和 TPT 等最先进方法取得了更好的识别性能。特别是，当五频带的 DE 特征组合在一起时，在受试者相关实验情况下，DGCNN 的平均识别准确率可高达 90.40%，而在受试者无关交叉验证实验情况下，平均识别准确率可高达 79.95%。在 DREAMER 数据库上，使用所提出的 DGCNN，效价、唤醒度和支配度的平均准确率分别为 86.23、84.54 和 85.02%，这些数值均高于 SVM、GraphSLDA 和 GSCCA。DGCNN 识别性能更好的原因最可能是以下主要因素：DGCNN 使用了非线性神经网络

使其在处理非线性判别特征学习方面更加强大；

DGCNN 的图表示提供了一种有用的方法来表征各种脑电图（EEG）通道之间的内在关系，这对于提取情绪识别任务中最具判别性的特征非常有利；DGCNN 通过优化邻接矩阵 W 自适应地学习 EEG 通道的内在关系。相比之下，GCNN 模型在模型学习阶段之前确定了 W 的值。因此，使用 DGCNN 模型比 GCNN 更准确地表征 EEG 通道之间的关系。此外，值得注意的是，邻接矩阵的对角元素表示 EEG 通道对 EEG 情绪识别的贡献。因此，邻接矩阵将提供一种潜在的方法来找出在 EEG 情绪识别中最具贡献的 EEG 通道，这将有利于进一步提高 EEG 情绪识别性能。

我们将这个有趣的主题留作未来的工作。

另一个未来可以研究的有趣问题是关于数据规模泛化。尽管所提出的 DGCNN 方法在处理脑电图情绪识别方面已证明是一种好方法，但值得注意的是，实验中使用的脑电图数据库的数据规模仍然相对较小，这不足以学习更强大的深度神经网络模型，因此可能会限制本研究的进一步性能提升。因此，需要一个规模更大的脑电图数据库来解决这个问题，这也是我们未来工作的主要任务。

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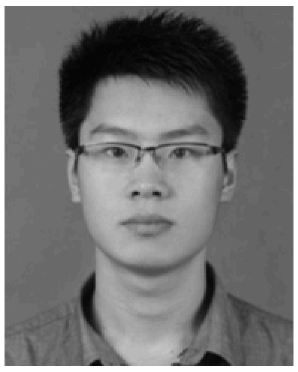

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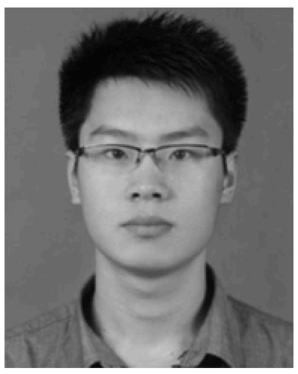

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